MATHS I · WEEK 4

Maths I week 4 formulas: Polynomials and their graphs

Polynomials: algebra & division · Graphs of polynomials. Save this PDF for revision.

POLYNOMIALS: ALGEBRA & DIVISION · OFTEN ASKED

Polynomials: algebra & division

Degree

highest power with a non-zero coefficient

Degree rules

$\deg(pq) = \deg p + \deg q \cdot \deg(p \pm q) \le \max(\deg p, \deg q)$

Division algorithm

$p(x) = d(x) \cdot q(x) + r(x), \deg r < \deg d$

Remainder theorem

$p(x) \div (x - a)$ leaves remainder $p(a)$

$\div (ax - b)$ leaves $p(b/a)$

POLYNOMIALS: ALGEBRA & DIVISION · OFTEN ASKED

Polynomials: algebra & division

Factor theorem

$(x - a)$ is a factor $\Leftrightarrow p(a) = 0$

Sum of coefficients

$p(1)$

Alternating sum: $p(-1)$

Rational root test

possible roots = ±(factors of constant)/(factors of leading coefficient)

Cubic $ax^3 + bx^2 + cx + d$

$\alpha + \beta + \gamma = -b/a$, $\alpha\beta + \beta\gamma + \gamma\alpha = c/a$, $\alpha\beta\gamma = -d/a$

POLYNOMIALS: ALGEBRA & DIVISION · OFTEN ASKED

Identities

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference of squares

$$a^2 - b^2 = (a + b)(a - b)$$

Cube

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

Sum / difference of cubes

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

GRAPHS OF POLYNOMIALS · IN EVERY PAPER

Graphs of polynomials

Real roots

degree n ⇒ at most n x-intercepts

Turning points

at most n – 1

Odd degree

at least 1 real root; the two ends point opposite ways

Even degree

both ends point the same way; may have no real root

Factor form

$p(x) = a(x - r_1)^{k_1}(x - r_2)^{k_2}\cdots$, degree = $k_1 + k_2 + \cdots$

y-intercept

p(0)

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End behaviour (leading term $a_{99}^{20}x^n$)

	$a_{99}^{20} > 0$	$a_{99}^{20} < 0$
n even	↑ … ↑ (both ends up)	↓ … ↓ (both ends down)
n odd	↓ … ↑ (down-left, up-right)	↑ … ↓ (up-left, down-right)

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Multiplicity at a root r

Factor	At x = r the graph
$(x - r)^1$	crosses straight through
$(x - r)^2$, any even power	touches the axis and bounces back (no sign change)
$(x - r)^3$, odd power ≥ 3	crosses but flattens out (S-shape)

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Key points

- Reading a graph: a crossing root has odd multiplicity; a touching root has even multiplicity (usually 2).
- Minimum degree from a graph = sum of the multiplicities you can see, and at least (turning points + 1).
- Find the leading coefficient a by substituting the y-intercept or any labelled point.
- p(x) changes sign only at odd-multiplicity roots. Use a sign chart for 'p(x) > 0 when?' questions.

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Common traps

⚠ 'Distinct roots' counts each root once. Counted with multiplicity, $(x - 1)^2$ is two roots.

⚠ Turning points aren't roots: x^3 has 1 root and 0 turning points; $(x + 2)(x - 1)^2$ has 2 distinct roots and 2 turning points.

⚠ At an even-multiplicity root the graph does not change sign.

Notes for every week of Maths I, free.

Full notes, the formula sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/maths-1 →

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